KAYAN INNOVATION LLC

شركة كيان للابتكار وريادة الأعمال وبيئات العمل

TECHNOLOGY OPERATING SYSTEM

نظام تشغيل التقنية

KAY-TECH-OS-008 · Version 8.0 · FINAL

Every decision made. Every system specified. Every screen described.

The complete technical blueprint for building Kayan from zero to globally recognised.

Commissioned: 20 April 2026 · Effective: 1 May 2026 · Owner: CEO + Dev Lead

Stack locked: Next.js 14 + NestJS 10 + PostgreSQL 16 + Redis + Hetzner CX31

Classification: CONFIDENTIAL — Dev Team + Board Only

Table of Contents

A · Why This Document Exists and How to Use It

This is not a wish list. Every decision in it has already been made.

This document is the single source of truth for every technology decision Kayan has made. It tells the development team exactly what to build, why it was designed that way, and how each piece connects to every other piece. It tells the CEO what has been decided and what the rationale was. It tells a future CTO what they are inheriting and why the architecture is the way it is.

The document is organised into eight parts. Part A explains the reasoning framework. Part B locks the technology stack with the rationale for every choice. Part C describes the complete site architecture — every subdomain, every page, every user journey, every click. Part D specifies the platform engine in full — verification, matching, pods, escrow, the e-working space. Part E describes the Kayan Card system. Part F covers the ERP, accounting, and automation layer. Part G is the infrastructure, security, and deployment specification. Part H is the build sequencing and accountability plan.

Any engineer reading this document should be able to understand what Kayan is, make correct implementation decisions without asking the CEO, and build the platform in a way that will not need to be torn down and rebuilt eighteen months later. That is the standard this document is written to.

A.1 The One Rule That Governs Every Technical Decision

> **No SOW + No Escrow (100% funded) = No Work.**
>
> *This is not a policy preference. It is hardcoded at the database level. No user — including admin or CEO — can bypass it. Every piece of technology in this platform exists to enforce and support this rule.*

Kayan is not a marketplace. It is a Verified Talent Map and Outcome Delivery Engine. The platform exists to solve three structural problems that destroy the talent economy in underserved markets: the verification gap (no one can prove who can actually do what), the governance gap (no structure for how work gets done and approved), and the trust infrastructure gap (no safe way for money to move). Every technology decision traces back to solving one or more of these three problems.

The result is that Kayan's technology must do things most platforms never attempt: verify identity under wartime conditions without reliable civil registry access, operate on 2 Mbps 3G connections without data loss, enforce financial controls that a CBY-Aden regulator could audit, and present a bilingual interface where Arabic is genuinely first-class — authored, not translated, displayed right-to-left without a single layout break.

B · The Locked Technology Stack — Every Decision and Its Rationale

ADR-001 is closed. The stack is locked. What follows is the reasoning so no one revisits it.

Architecture Decision Record 001 was ratified on 19 April 2026. The Laravel/PHP proposal that was circulating in earlier drafts of the Tech OS was reviewed and rejected. The rejection had three grounds. First, the existing development team's expertise is in TypeScript and Node.js, and switching stacks mid-project would cost four to six weeks of ramp-up time that the June 1 launch cannot absorb. Second, an apples-to-apples capability comparison showed NestJS delivering equivalent functionality with better long-term maintainability for a team of this size. Third, the TypeScript-first approach means the same type definitions and validation schemas can be shared between the frontend and backend, eliminating an entire class of integration bugs that are common in polyglot stacks. ADR-001 is closed.

B.1 The Full Stack, Layer by Layer

Layer	Technology and Rationale
Frontend	Next.js 14 with the App Router, TypeScript, and Tailwind CSS. Chosen because server-side rendering is non-negotiable for Arabic SEO, the App Router enables per-route streaming which matters on slow connections, and the TypeScript ecosystem aligns with the backend. Every subdomain runs on the same Next.js codebase deployed to Hetzner, edge-cached via Cloudflare Workers. Tailwind is used for utility-first styling that survives RTL layout without the class conflicts that CSS Modules create in Arabic interfaces.
Backend API	NestJS 10 with TypeScript. Chosen for its modular, decorator-based architecture that enforces separation of concerns in a way that scales as the team grows. Each of Kayan's nine operational queues maps cleanly to a NestJS module. The built-in dependency injection makes testing straightforward. The REST API handles all client-server data transfer; WebSocket via Socket.io handles real-time workspace events. GraphQL was considered and rejected: the extra schema layer adds complexity that is not justified at this scale, and REST is easier for the external API consumers (enterprise clients, INGO procurement systems) who will use the API in Phase 3.
Database	PostgreSQL 16 with the pgvector extension and Prisma as the ORM. PostgreSQL is the only choice that gives Kayan all of the following simultaneously: ACID compliance for financial data, full-text search in Arabic and English, vector similarity search via pgvector for talent matching, row-level security for RBAC

Layer	Technology and Rationale
	enforcement, and a mature extension ecosystem. Prisma provides type-safe query building and database migration management. The pgvector extension enables semantic talent-project matching using Cohere multilingual embeddings stored directly in the database — no separate vector database needed at this scale.
Cache and Queues	Redis with BullMQ. Redis handles session storage, rate limiting counters, and real-time presence indicators. BullMQ provides the nine operational queues (verification, matching, intake, delivery oversight, QA gate, payout approvals, disputes, trust and safety, retention) with retry logic, dead-letter queues, and job priority. Redis is deployed as a single instance on Hetzner alongside the main application; a Redis Sentinel pair will be added when the Series A funds Phase 2 infrastructure.
Real-time	Socket.io with a Redis adapter. Socket.io handles live workspace events: gate transitions, new messages in the project channel, escrow state changes, and verification status updates. The Redis adapter ensures that Socket.io events broadcast correctly across multiple Node.js instances as Hetzner scales horizontally. WebSockets are only active when a user has a workspace open — not on public pages, not in the admin console — to minimise battery drain on mobile devices in Yemen.
Object Storage	MinIO self-hosted on Hetzner for non-sensitive assets (project deliverables, card designs, profile photos, SOW PDFs). AWS S3 in the Bahrain region with KMS encryption for sensitive data (KYC documents, identity photographs, liveness videos, biometric hashes). The split is deliberate: MinIO has no data residency concerns and zero per-request cost; S3 Bahrain keeps biometric data in a region that is closer to Yemen and outside the major jurisdictions that could issue legal process for user data without notice.
Authentication	Auth.js v5 with magic-link email as the primary flow and SMS OTP as the fallback. Password-based authentication is not offered. Password reset is a notoriously poor user experience and a common attack vector; magic-link is more secure and simpler for a population where many users share devices. TOTP-based MFA is required for all admin and finance users and

Layer	Technology and Rationale
	optional but encouraged for clients with escrow funds. Session duration is 14 days for talent and clients with activity extension, 8 hours for admin and staff.
AI — SOW Generation	Anthropic Claude Sonnet 4.5 via the API. Claude drafts the bilingual SOW within 60 seconds of a client submitting a project brief. The prompt includes the full brief, the taxonomy of the matched micro-specialization, the milestone structure template for that project type, and the Kayan SOW schema. Claude outputs structured JSON that the platform renders as a formatted bilingual PDF. Claude also summarises dispute evidence packages for the Arabic arbitrator. The API cost is approximately $150 per month at Year 1 volume.
AI — Matching	Cohere embed-v4 multilingual for semantic embedding and Anthropic Claude Haiku for reranking. When a project brief is submitted, the platform embeds the brief using Cohere, queries pgvector for the top 50 talent profiles by cosine similarity, then passes those 50 profiles and the brief to Haiku with a reranking prompt that weights verified delivery history, axis scores, and team compatibility signals. Haiku outputs an ordered shortlist with a reasoning snippet for each recommendation. Cohere was chosen over OpenAI Embeddings specifically because its multilingual model handles Arabic and English mixed-language documents without the quality degradation that English-first models show.
AI — Arabic STT	Intella for Arabic dialect speech-to-text. Used for voice-input brief capture on mobile — a critical accessibility feature for users who can speak their project requirement more fluently than they can type it. Intella was chosen specifically for Yemeni and Gulf Arabic dialect support. Google Cloud STT and Amazon Transcribe both underperform on Yemeni dialect; Intella is purpose-built for the MENA market.
KYC / Identity	Persona Startup Program for Year 1 (free). Persona handles government ID extraction, ICAO-compliant face match, and liveness detection via an SDK embedded in the verification flow. At Year 1 volumes (under 200 verifications per month) the Persona Startup tier is free. The platform architecture is designed to swap Persona for Sumsub in Year 2 without changing the

Layer	Technology and Rationale
	verification flow — the vendor is abstracted behind a KYC service interface.
Sanctions Screening	ComplyAdvantage Starter at $49/month. Screens against UN Security Council Resolutions 2140 and 2216 (Yemen-specific), OFAC SDN, EU consolidated, UK OFSI, and PEP lists. Every new user (talent, client, vendor) is screened at registration and re-screened at contract initiation. Ongoing watchlist monitoring triggers an alert within 24 hours of a new match. ComplyAdvantage was chosen over free OpenSanctions because it provides the automated monitoring and the structured API response format that feeds into the Kayan verification audit trail.
Payments (intake)	Stripe via the Kayan FZ-LLC UAE merchant account. Stripe handles all client payment intake: card payments, ACH where applicable, and Apple/Google Pay on mobile. The Kayan FZ-LLC UAE entity provides the legal and banking credibility that international clients and Gulf businesses require. Stripe was chosen over Checkout.com as the primary rail because of its superior developer experience, SDK maturity, and fraud detection tooling. Checkout.com remains on the shortlist as a backup rail.
Payouts (talent)	InstaPay via the Kayan Misr Egypt entity for Egyptian contractors. Aden Bank Yemen for Yemeni talent (manual transfer while Tadamon integration is in progress). USDT via a regulated exchange as a USD-equivalent fallback rail for talent who cannot receive via banking. The payout architecture is multi-rail by design: no single payout failure should block a talent member from being paid.
Hub Management	Spacebring at $129/month. Manages desk and room booking, member access credentials, occupancy reporting, and the self-service member portal at space.kayanwork.com. Spacebring was chosen over Nexudus and OfficeRnD because it has the best Arabic localisation support of the three and its API is well-documented for the Kayan platform integration.
Café POS	Loyverse at $25/month. Handles sales recording, inventory tracking, shift reporting, and daily cash-up export. The export

Layer	Technology and Rationale
	feeds directly into KAYFIN_v5 PETTY_CASH sheet via a CSV import. Loyverse was chosen for its offline mode — critical for Aden power outages — and its Arabic interface.
Kayan Forge LMS	Moodle hosted on Hetzner at $25/month for a managed instance. Handles cohort coursework, assignment submission, grade tracking, and completion certificates. The Kayan Forge completion trigger in Moodle fires a webhook to the Kayan API, which updates the talent's Track Record axis score and queues a KC-A card issuance request. Moodle was chosen for its full open-source auditability, strong Arabic support, and the ability to customise the completion certificate to embed the Kayan brand and signed Open Badge 3.0 metadata.
Hosting	Hetzner CX31 in Falkenstein, Germany at €14.96/month. Runs all application containers (Next.js, NestJS, Redis, MinIO) via Docker Compose. The Falkenstein location was chosen over Helsinki and Ashburn for latency to Aden: Falkenstein sits on the DE-CIX Frankfurt exchange, which has the best peering to the Middle East submarine cable infrastructure. Cold standby backups go to Hetzner's Helsinki DC. The CX31 (4 vCPU, 8 GB RAM) handles Year 1 load comfortably. Migration to Hetzner CX41 (8 vCPU, 16 GB RAM) is planned at 200 concurrent active users.
Edge / CDN / WAF	Cloudflare Pro at $20/month. Provides DNS management for all 14 subdomains, WAF rules tuned to OWASP Top 10, DDoS protection, edge caching of static assets and ISR pages, and Cloudflare Workers for edge-side language detection and session token validation. The language detection at the edge — defaulting Arabic for users on Yemen/MENA IP ranges — eliminates a round-trip to the origin server for the most common case.
Monitoring	Uptime Kuma self-hosted on Hetzner for uptime monitoring across all 14 subdomains, with alerts to Slack and WhatsApp. Grafana with the Prometheus exporter for application-level metrics. Both are free and run on the same Hetzner CX31 at negligible resource cost. status.kayanwork.com is driven by the Uptime Kuma API.

Layer	Technology and Rationale
Error Tracking	Sentry free tier for frontend and backend error tracking. Every unhandled exception in Next.js and NestJS is captured with full stack trace, user context, and request payload. Critical errors page the on-call person via WhatsApp within 5 minutes.
Email	Postmark for transactional email (magic-link, verification notifications, SOW delivery, payout confirmations). Postmark was chosen for its delivery rate and its clear template system that supports bilingual Arabic/English email bodies. Google Workspace handles internal staff email at $6/user/month.
CI/CD	GitHub Actions for automated testing and deployment. On every push to main: run Prisma migrations on a test database, run the NestJS test suite, run the Next.js build, and if all pass, deploy to Hetzner via SSH. The entire pipeline runs in under 8 minutes. Zero manual deployment steps after the initial server setup. Code ownership: all source code in a GitHub organisation owned by the Kayan Innovation LLC account, with Maher as the organisation owner. No contractor has organisation owner access.

B.2 Monthly Technology Cost Breakdown

The total monthly technology cost at the locked stack is $1,075. This is within the $1,200 constitutional sub-ceiling and has $125 of headroom for unexpected cost increases.

Service	Monthly Cost	Why this and not something cheaper
Hetzner CX31	$24	The cheapest server that handles Year 1 load. Comparable DigitalOcean droplet costs $48.
Cloudflare Pro	$20	Free tier lacks WAF rules. WAF is non-optional for a platform handling money.
AWS S3 Bahrain + KMS	$50	Biometric data cannot sit on the same server as application code. No cheaper compliant

Service	Monthly Cost	Why this and not something cheaper
		option.
Persona Startup (KYC)	$0	Free for Year 1. Switches to ~$200/mo at Year 2 volumes.
ComplyAdvantage	$49	No adequate free sanctions screening API with automated monitoring exists.
Claude API (SOW + disputes)	$150	Estimated at Year 1 volume. Claude Haiku for matching is cheaper; Sonnet for SOW drafting.
Cohere embed-v4	$40	Arabic-English multilingual embeddings. OpenAI degrades on Arabic.
Intella Arabic STT	$30	Only viable Arabic dialect STT for Yemen. No alternative at this price.
Moodle (hosted)	$25	Kayan Forge LMS with custom certificate and webhook support.
Spacebring (Hub)	$129	Best Arabic-localised coworking software. Required for Hub operations.
Loyverse (Café POS)	$25	Offline mode. Arabic interface. Direct CSV export for accounting.
Domain + SSL	$15	Annualised across all 14 subdomains.

Service	Monthly Cost	Why this and not something cheaper
Monitoring (Uptime Kuma + Grafana)	$20	Self-hosted on Hetzner. Negligible resource cost.
Postmark email	$10	Transactional email. Better delivery rate than SendGrid free tier.
Tech contingency	$98	Buffer to the $1,200 ceiling. Absorbs Intella overages and Stripe webhook costs.
TOTAL	$1,075	$125 headroom within the $1,200 constitutional sub-ceiling.

C · The Complete Site Architecture

Fourteen subdomains. Every page specified. Every user journey mapped.

Kayan runs on fourteen subdomains under kayanwork.com. This is not over-engineering. The subdomain structure reflects genuine operational boundaries: different audiences, different RBAC requirements, different caching strategies, and in some cases different regulatory risk profiles. The edge layer (Cloudflare Workers) handles subdomain routing, language detection, and session validation before any request reaches the Hetzner origin.

C.1 Subdomain Map

Subdomain	Audience	Phase	Go-Live
kayanwork.com	Everyone — brand, trust, first contact	Phase 1	1 May 2026
platform.kayanwork.com/client	Business clients posting projects	Phase 1	30 June 2026
platform.kayanwork.com/talent	Verified freelancers doing work	Phase 1	30 June 2026
space.kayanwork.com	Hub members and visitors	Phase 1	30 June 2026
forge.kayanwork.com	Learners, instructors, cohorts	Phase 1	30 June 2026
volunteers.kayanwork.com	Kayan for Yemen volunteers	Phase 0	25 April 2026
impact.kayanwork.com	UN/INGO, donors, press	Phase 1	30 June 2026
card.kayanwork.com	Public card verification, cardholders	Phase 1	30 June 2026
yeto.kayanwork.com	Clients buying fixed-scope deliverables	Phase 2	Q3 2026
admin.kayanwork.com	Kayan staff — operations, escrow, verification	Phase 1	30 June 2026

Subdomain	Audience	Phase	Go-Live
staff.kayanwork.com	All Kayan employees — daily work portal	Phase 1	30 June 2026
api.kayanwork.com/v1	Enterprise API consumers	Phase 3	Q4 2026
docs.kayanwork.com	Developers, power users	Phase 3	Q4 2026
status.kayanwork.com	All users — platform health	Phase 1	30 June 2026

C.2 kayanwork.com — The Brand Surface

This is where the world first meets Kayan. Every word, every image, and every interaction on this domain has one job: converting a skeptic into someone who believes Kayan is real, serious, and different from everything else in this market.

The homepage is not a brochure. It is a trust machine. The design is calm — a deep oud-green near-black gradient background, gold accent elements, and typography that prioritises readability over decoration. The hero line reads: "Verified Teams. Governed Delivery. Outcomes you can trust." This is not a tagline generated by a marketing agency. It is an accurate description of what the platform actually does. Under it: "Verified talent map and managed delivery workspace." Then three equal entry doors: I need a result (clients), I want to work (talent), I oversee delivery (internal and partners).

The three doors are visually equal. This is a deliberate product decision. On every other freelance platform, the client is primary and the talent is secondary — "post a job," then talent applies. Kayan rejects this framing. Talent is not applicants. They are verified professionals. The platform serves both sides with equal dignity, and the homepage design enforces that before a single word is read.

Page structure for kayanwork.com: Homepage, About, Manifesto (the ten public commitments, signed and dated), Team (real names, real photos, no stock), The Verified Standard (what verification means, publicly explained), Press, Careers, and Contact. Six legal pages at /legal/ (Terms, Privacy, Cookies, AML-KYC, Acceptable Use, Complaints). Each legal page is bilingual full-text with a permanent URL and downloadable as PDF.

C.3 The User Journey: A First-Time Client

Understanding exactly what a first-time client experiences — from the moment they land on kayanwork.com to the moment their first project is delivered — is the most important thing a developer can know before writing a single line of code. What follows is that journey, described as it actually happens.

A procurement officer at an INGO in Nairobi hears about Kayan from a colleague who used it for a data analysis project in Aden. She searches "Kayan work Yemen" and lands on kayanwork.com. The page loads in under two seconds. She reads the hero line. She is skeptical — she has tried Upwork for Africa-based talent and found the verification claims hollow. She scrolls. She sees the Verified Standard section, which explains the five-axis grid in plain language: Identity, Skills, Track Record, Behaviour, Capacity. She reads that Identity verification requires a liveness check, not just an ID upload. She clicks "I need a result."

She arrives at platform.kayanwork.com/client. She is not yet logged in. The page shows a value proposition specifically calibrated for institutional buyers: escrow-gated delivery, bilingual contracts, audit trail for procurement compliance, and a sample project timeline with gate checkpoints. She clicks "Post a brief." She is asked to create an account. The signup form asks for name, email, and organisation. Magic link sent. She clicks the link. She is in.

She fills in the brief wizard. Step one: project type (she selects "Data analysis and visualisation"). Step two: what is the business goal ("Produce a household income survey analysis for our Yemen programme"). Step three: constraints (deadline 3 weeks, language Arabic and English, data is survey CSV files, confidentiality required). Step four: budget range ($1,500 to $3,000). Step five: she selects "Sensitive" from the security tier options, because the survey data contains personally identifiable information. She submits.

Within 60 seconds, Claude has drafted a bilingual SOW. It appears on screen: scope, deliverables, acceptance criteria, milestone plan (Milestone 1: data cleaning and exploration report; Milestone 2: analysis and visualisation package; Milestone 3: final bilingual report and presentation deck), review windows, and the escrow structure. The 2170 split is shown explicitly: 70% to contractor, 9% to Kayan, 21% held for 90 days. She reviews and approves with one click.

The matching engine has already been running. While she was reviewing the SOW, the platform embedded her brief, queried pgvector for the top 50 talent profiles by cosine similarity, and passed the shortlist to Claude Haiku for reranking. She sees a proposed pod: a T3 Gold data analyst (Pod Lead, with a project on household survey analysis in her prior track record) and a T2 Silver data visualisation specialist. Each profile shows the five-axis grid, the verification tier, a reasoning snippet explaining why this person was selected, and a performance summary: 12 projects, 94% on-time, 0 disputes, 4.8 average rating. She approves the pod with one click.

She is taken to the escrow funding screen. Milestone 1 is $800. She funds it via card through Stripe (Kayan FZ UAE merchant). The 2170 decomposition applies: $560 (70%) enters the contractor settlement queue, $168 (21%) enters the 90-day reserve, $72 (9%) is recorded as Kayan fee revenue. She receives a bilingual payment receipt and is taken to the e-working space for her project.

Over the next three weeks, she reviews progress in the workspace. Every decision is in the decision log. Every file is versioned. The QA reviewer has checked the G6 interim deliverable against the acceptance criteria before it reached her. She approves each milestone with one click. When the final deliverable arrives, it matches exactly what was described in the SOW she approved three weeks ago. She approves the final milestone. The escrow releases. She rates the pod. The platform suggests she keep this pod for future projects. She says yes.

That is the experience Kayan must deliver. Every line of code exists to make that experience possible. The developer who builds the brief wizard, the matching endpoint, the SOW generation prompt, or the workspace milestone module should read this journey and understand exactly what they are building toward.

C.4 The User Journey: A First-Time Talent Member

A graphic designer in Aden, 24 years old, hears about Kayan at an event at the Hub. She has done freelance work before — mostly through informal WhatsApp referrals — and has never been paid reliably. She visits kayanwork.com on her phone.

She clicks "I want to work." She lands on platform.kayanwork.com/talent. She reads the verification ladder: T0 is just a profile, T1 is identity verified, T2 is when she can join paid projects. She can see exactly what it takes to get to T2: identity verified, skills assessment passed, one monitored project completed. She clicks "Start registration."

Registration takes five minutes. She enters her name (four-part Arabic name: given, father, grandfather, family), phone, email, and selects her micro-specializations: Logo and Brand Identity (primary), Social Media Design (secondary), UI/UX Design (secondary). The form shows an example of "strong evidence" for each specialization to set her expectations. She uploads two portfolio pieces with provenance notes — for each piece she says what she did, for whom, what the brief was, and whether she can provide a client contact. She submits.

She receives a confirmation that her application is in the verification queue. Within 48 hours, she receives a WhatsApp message from the Verification Officer at the Aden Hub inviting her to book a verification appointment. She books for the following Saturday at 10 AM.

She arrives at the Hub. She is offered Yemeni coffee. The Verification Officer walks her through the process: government ID photograph (front and back, photographed against the white wall with the lighting guide on screen), ICAO-standard selfie, the ten-second liveness challenge (head turn, blink). The Persona SDK handles the face match and ID extraction in under two minutes. Then a 90-minute skills assessment on a Hub laptop: a real design brief, executed in Adobe Illustrator, reviewed against the scoring rubric for Logo and Brand Identity. Then a 30-minute structured interview. The Officer scores each axis on the 1-5 rubric. Total time: 2.5 hours. She is told she will receive a decision within 7 business days.

Five days later: approved at T2 Silver. Her profile goes live on the platform. The KC-T Silver card is in the next print batch. Within the week, she receives her first project invitation: a branding project for a Yemeni startup client, matched to her Logo and Brand Identity specialization, $650 escrow, 14-day timeline. She accepts. She joins the project workspace. She delivers. The client approves. $455 (70%) releases to her InstaPay account. She has just done her first formally verified, escrow-backed, auditable piece of work. Everything is documented. The track record has begun.

C.5 The User Journey: A Kayan for Yemen Volunteer

A 20-year-old university student in Aden sees the Kayan for Yemen call on Instagram. The post shows @kayanhub and says: "Be part of something that matters. Become كيان لليمن. Apply now." She visits volunteers.kayanwork.com on her phone. The page explains Kayan for Yemen in two paragraphs: what it is, what volunteers do, what they get. Eight role cards. She selects Community Ambassador (primary) and Translation (secondary). The application form takes 4 minutes. She submits.

The Kayan for Yemen Coordinator calls her within 48 hours. She is scheduled for the PSEA orientation session the following Wednesday at the Hub. She attends. She signs the consent form. She receives her KC-V Volunteer Card: sage green, her name in Arabic and English, her photo, a QR code, the word كيان لليمن embossed under the Kayan logo. She is told her first shift is Saturday morning, Hub reception.

Over three months, she logs 42 hours. Her 40-hour milestone triggers an automatic notification: "You have earned T1 Talent fast-track status. Would you like to begin your talent verification?" She says yes. Her volunteer hours are recognised as demonstrated community experience. The identity verification is scheduled. The skill assessment for Translation (Arabic-English) is waived for the identity-only T1 path; she will need to do the skills assessment for T2 separately. But she is on the platform now. Her path from volunteer to verified paid talent is mapped, and she can see every step.

D · The Platform Engine — Verification, Matching, Pods, Escrow, and the E-Working Space

D.1 The Verification System — T0 to T4

Kayan's verification is the core product. Everything else — the matching, the escrow, the governance — depends on the verification system being trustworthy. The system uses five axes, each scored 0 to 5. The composite tier is the minimum of all five axes — a talent member cannot be T3 if any single axis is below L3.

Axis	What it measures and how
I — Identity	Government ID extraction via Persona SDK (front + back + liveness video). ICAO-compliant face match. For applicants without current government ID (common in a conflict zone): tribal/Sheikh attestation or family book combined with a head-of-household ID, capped at L1-L2. Result: a confirmed match between the person physically present at the Hub and the identity record in the database. No digital-only path for T2+.
S — Skills	Domain-appropriate task-based assessment calibrated per micro-specialization. Graphic design: a real brief executed in Adobe Illustrator, scored by a T3+ reviewer on the QA rubric (Completeness 25%, Quality 30%, Functionality 20%, Brand Alignment 10%, Documentation 15%). Development: a 2-3 hour coding challenge with code review. Translation: a live translation under timed conditions. The assessment is on a Hub laptop — no "bring your own device" for the assessment stage, to prevent cheating. Kayan Forge cohort completion automatically upgrades the S axis to at least L2 on the trained specialization.
T — Track Record	Performance ledger: on-time delivery rate, revision rate, client acceptance rate, rehire rate. For new talent: starts at L1. Increases with each completed project. Immutable: entries can only be added, never edited. Admin-level audit event required for any correction.
B — Behaviour	Communication responsiveness (response time within the workspace SLA), missed deadline rate, dispute frequency, professionalism ratings from clients and pod leads. 90-day rolling window. Two consecutive 90-day cycles below the tier threshold triggers automatic demotion with a repair path — never silent removal.

Axis	What it measures and how
C — Capacity	Honest weekly capacity declaration. Anti-double-booking enforcement. Actual utilisation vs declared availability. If a talent member is consistently unavailable despite declared availability, the C axis degrades and a system notification is sent.

The tier structure governs what each talent member can access on the platform. This is a permissions system, not a badge system. The difference matters: permissions have hard enforcement; badges are cosmetic. T0 can complete a profile but cannot apply for any project. T1 can observe in pods and take low-risk internal tasks. T2 can participate in all commercial projects as a Junior role. T3 can take lead roles. T4 is Pod Lead eligible.

Tier	Access rights and benefits
T0 — Registered	Profile only. No project access. No card issued. Visible in talent directory as "pending verification." Cannot be contacted by clients.
T1 — Identity Verified	Observer role in pods. Low-risk internal projects. Day pass at Hub standard rate. KC-T Basic card issued (digital only, grey band). Kayan for Yemen fast-track route available at 40 volunteer hours.
T2 — Skills Verified (Silver)	Full commercial platform access. Junior pod roles. KC-T Silver card issued (physical card, silver band). 10% discount on Hub memberships and Café. Featured in client shortlists. Eligible for Kayan Forge Intermediate cohort.
T3 — Track Record (Gold)	Lead roles in pods. QA Reviewer role. Enterprise project eligibility. KC-T Gold card (gold band). 20% discount on Hub and Café. Access to monthly T3+ networking event. Featured first in matching recommendations.
T4 — Proven Performer (Platinum)	Pod Lead role. All project types including regulated. INGO procurement projects. Platinum card (dark metallic band). Unlimited hot desk included in membership. Invitation to T4 Expert Panel (advisory role, honorary). Priority on Series A referral programme.

D.2 Micro-Specializations — The Identity Layer

Kayan does not use generic job titles. It uses micro-specializations — specific, testable, matchable capability claims. A talent member is not "a designer." They are "a Logo and Brand Identity specialist" or "a Social Media Design specialist" or "a UI/UX Design specialist." Each micro-specialization has its own assessment rubric, its own portfolio requirements, and its own client-visible evidence standard.

The platform launches with 50 micro-specializations across five domains. This number is deliberate: broad enough to cover the Aden talent pool, narrow enough that each specialization has a meaningful verification standard. New micro-specializations are added quarterly, based on client demand signals from the intake queue.

Domain	Micro-Specializations	Demand in Aden Market
Design	Logo and Brand Identity · Social Media Design · UI/UX Design · Print and Packaging · Motion Graphics · Illustration	Very High
Development	WordPress Sites · Mobile Apps (React Native) · Web Applications · E-Commerce (WooCommerce/Shopify) · API Integration · DevOps and Infrastructure	Medium-High
Content and Media	Arabic Copywriting · Arabic-English Translation · Video Production and Editing · Photography (Commercial) · Podcast Production · Arabic SEO	Very High
Marketing	Social Media Management · Paid Advertising (Meta/Google) · Email Marketing · SEO and Content Strategy · Market Research · PR and Media Relations	High
Business and Consulting	Business Planning and Feasibility · Financial Modelling · HR Policy and Organisational Design · Legal Research and Contract Drafting · Project	Medium

Domain	Micro-Specializations	Demand in Aden Market
	Management · Data Analysis and Visualisation	

Each micro-specialization maps to a specific assessment protocol in the verification SOP (KAY-HR-014). The assessment protocol specifies: the task brief for the skills assessment, the time limit, the tools available, the scoring rubric dimensions and weights, the minimum passing score, and the number of portfolio pieces required at each tier level. No micro-specialization may be listed on a talent profile unless it has been assessed through this protocol.

D.3 The Matching Engine

When a client submits a project brief, the matching engine runs automatically. The process has four stages: embedding, retrieval, reranking, and pod composition.

In the embedding stage, the brief text (Arabic and English combined) is passed to the Cohere embed-v4 multilingual API. The result is a 1024-dimensional vector. This vector is stored in PostgreSQL alongside the project record using the pgvector extension. Simultaneously, every verified talent profile maintains a pre-computed embedding of their combined profile text (specializations, portfolio descriptions, work history), updated whenever the profile changes.

In the retrieval stage, pgvector performs a cosine similarity search between the brief embedding and the talent profile embeddings. The top 50 candidates by similarity score are retrieved. The query is fast: pgvector with an IVFFlat index on the profiles table returns results in under 50 milliseconds even with 5,000 profiles.

In the reranking stage, the 50 candidates and the brief are passed to Claude Haiku with a structured prompt. The prompt instructs Haiku to rerank the candidates considering: axis scores (weighted by project risk level), verified track record in similar project types, current availability, team compatibility history with other candidates in the shortlist, and linguistic match (Arabic-primary clients matched with Arabic-fluent talent first). Haiku outputs a JSON array of ranked candidates with a reasoning snippet for each.

In the pod composition stage, the top candidates per required role are grouped into 2-3 pod configurations. Each configuration includes a risk score, an estimated timeline confidence rating, and a pricing guidance suggestion. The human Delivery Lead reviews the configurations and makes the final selection. The Delivery Lead's override is logged as a training signal — if their selection consistently outperforms the algorithmic recommendation, the reranking weights are updated.

D.4 The Pod Assembly Protocol

A pod is not a permanent team. It is a project-specific assembly of verified talent. Every pod has a defined structure. The Pod Lead is T4-verified, accountable for delivery and client liaison, the only person who can approve a deliverable for client review, and the person who signs the closeout checklist. Specialists 1 to 3 are T2+ verified in the relevant micro-specializations. The QA Reviewer is T3+ in the domain, has not been involved in the project execution, and reviews against the acceptance criteria before the client sees anything. The Support/Junior role is T1+ and handles documentation, asset preparation, and testing.

No pod starts work until three conditions are all true simultaneously: the SOW has been signed by both parties, the escrow has been funded for at least Milestone 1, and every pod member has confirmed acceptance of their role. These three conditions are enforced at the database level — the G4 to G5 transition query checks all three. If any is false, the gate does not open. This is the technical implementation of the foundational rule: No SOW + No Escrow = No Work.

The pod assembly is visible to the client. They see each team member's name, their tier badge, their verification summary, and the reasoning for their selection. This transparency is not optional — it is what makes the client's approval of the pod a genuine informed decision rather than a rubber stamp.

D.5 The G0 to G9 Gate System

Every project moves through ten gates. Each gate is a state in the project state machine. Each transition writes an event to the audit log. No gate can be bypassed. The gate system is how Kayan turns the informal chaos of project delivery into a governed, auditable, accountable process.

Gate	What happens and what triggers the next gate
G0 — Lead Intake	Client submits brief. System auto-classifies project type, complexity, and sensitivity level. Account Lead reviews and approves triage within 4 hours. Rejected briefs receive a bilingual explanation within 24 hours. Approved briefs move to G1.
G1 — Qualification	Client KYC checked. Sanctions screening fires. Intent deposit ($20-$50 refundable) funded. All checks pass: move to G2. Any check fails: brief held, client notified, human review triggered.
G2 — Scoping	Claude drafts bilingual SOW from brief. Delivery Lead reviews and adjusts. Client reviews and either approves or requests changes. This is the last point at which the scope can be substantially changed. Approved SOW is locked as a PDF in the project vault. Move to G3.
G3 — Pod Formation	Matching engine runs. Delivery Lead selects pod from 2-3 configurations. Each pod member is notified and must formally accept their role within 24 hours. All accept: move to G4. Any decline: replacement candidate offered. No pod can move forward with a role unconfirmed.
G4 — Escrow Funded	Client funds Milestone 1 via Stripe. 2170 decomposition applies immediately. System confirms funds received. Move to G5. Funding not received within 7 days: brief returns to G3 with a

Gate	What happens and what triggers the next gate
	client notification and an automatic escrow reminder sequence.
G5 — Kickoff	Three automated checks: SOW signed (confirmed at G2), escrow funded (confirmed at G4), all pod members confirmed (confirmed at G3). All three true: workspace activates, Loom kickoff video sent within 48 hours, project channel opened in the workspace. Any check false: system blocks and alerts Account Lead.
G6 — Interim Deliverable	Pod submits milestone deliverable. QA Reviewer applies the five-dimension rubric (Completeness 25%, Quality 30%, Functionality 20%, Brand Alignment 10%, Documentation 15%). Score ≥ 3.5 with no dimension below 3: deliverable goes to client review. Score 2.5-3.4: rework with specific feedback. Score < 2.5: escalate to CEO. Client sees deliverable only after QA pass. Client has a defined review window (typically 5 business days) to approve, request changes, or escalate.
G7 — Final Deliverable	Same QA process as G6. Additionally: handover checklist completed (all assets transferred, access credentials handed over, documentation complete). Client approves: move to G9. Client rejects: move to G8. The G7 handover checklist is a required artifact — no G9 without it.
G8 — Dispute	Client disputes the final deliverable. Escrow freezes. System compiles the evidence package: SOW, all deliverable artifacts, the decision log, the communication thread. Arabic arbitrator assigned. Evidence package and a Claude-generated neutral summary of both sides delivered to arbitrator within 48 hours. Decision within 7 days. Decision enforced through escrow: partial, full release, or full refund as determined.
G9 — Settlement	70% of escrow releases to contractor. 9% to Kayan operating account. 21% held in 90-day reserve. Invoices auto-generated for client and talent. Structured review request sent to both parties (5-dimension rubric, not star ratings). Operational memory updated: team fingerprint refined, template library updated, risk library updated. Continuity suggestion sent: "This pod delivered successfully. Would you like to reserve them for your next project?"

D.6 The E-Working Space

The e-working space is the product. Every other surface on the platform exists to get users into a workspace and keep them working within it. The workspace is what makes Kayan's governance model real — it brings all work, all decisions, all files, and all approvals into one auditable surface.

The workspace is organised into eight modules. These are not just UI sections — each is a distinct data domain with its own write access rules, its own retention policy, and its own audit event types.

Module	Specification
Project Dashboard	Single-screen status view: project name, security tier badge, current gate, escrow balance (funded/reserved/released), pod composition, outstanding actions, and a timeline showing all gates with their completion dates. On mobile: cards stack vertically with outstanding actions shown in a sticky banner. The dashboard refreshes in real-time via WebSocket when any gate event fires.
Milestones	Every milestone is a structured object with six required fields: deliverables list (what will be produced), acceptance criteria (how the client will judge completion), QA checklist (what the reviewer must verify), review window (how many business days the client has to respond), escrow amount, and assigned pod members per deliverable. No milestone can be created without all six fields. The milestone cannot be approved without all required artifacts attached. Artifact attachment is enforced at the database level, not just the UI.
Tasks	Kanban-style task management: Backlog, In Progress, In Review, Done. Each task has an owner (single person, not a group), a due date, dependencies on other tasks, and a tag mapping to the micro-specialization taxonomy. Task tags generate performance signals: completed tasks in the specialization contribute to the T axis score. Overdue tasks trigger an automatic alert to the Pod Lead. Overdue tasks that are 3+ days past due and unresolved trigger an alert to the Account Lead.
Files and Evidence Vault	Versioned file storage for all project assets. Naming convention enforced (project_milestone_version_date.ext). Files can be uploaded, previewed, and commented on. The "Final" designation can only be applied by the Pod Lead after client acceptance — it locks the file permanently. In Sensitive mode, file downloads are restricted to workspace members only and

Module	Specification
	every download is logged. In Regulated mode, downloads are disabled and files can only be viewed within the browser.
Communication	Three communication surfaces. The chat channel is a structured thread, not a free-form messenger — every message is associated with a gate or a milestone. The decision log is a separate surface for formal decisions: structured entries with a decision title, the options considered, the decision made, the rationale, and the timestamp. No decision about scope, budget, or timeline may be made in the chat channel; it must be in the decision log. The weekly checkpoint form is a mandatory structured update submitted every Monday by the Pod Lead showing: work completed last week, work planned this week, blockers, and any risks to the milestone timeline.
Approvals	The three-button interface. Approve: releases the milestone and triggers escrow release. Request Changes: returns the deliverable to the pod with a mandatory written explanation of what must change (minimum 50 characters — no one-word rejections). Escalate: opens the G8 dispute flow. There is no ambiguous state. Every client action is explicit. Silence after the review window closes triggers an automated reminder sequence (Day 1, Day 3, Day 5) before the Account Lead is notified.
Workspace Security Tiers	Standard: baseline audit logs, normal file access. Sensitive (PII): download restrictions, browser-only file preview, watermarking on all previews showing the viewer's name and timestamp, tighter role permissions (client cannot see contractor-internal working files). Regulated: mandatory NDA acceptance as a gate before workspace entry, 2FA enforced for all workspace sessions, all actions logged with enhanced detail, file downloads disabled. The security tier is shown as a badge in the workspace header on every page — not in settings, not in a modal, but always visible.
Audit Trail	Immutable, append-only log of every event in the project lifecycle. Events logged: file uploads and downloads, milestone submissions and approvals, gate transitions, escrow events, decision log entries, communication messages, and any admin override actions. Every event records: actor (user ID), timestamp (UTC), IP address, action type, and the entity affected. The audit

Module	Specification
	trail cannot be edited. Admin-level events (corrections, dispute outcomes) are logged as additional entries, not modifications of existing ones. The audit trail is exportable as a signed PDF for procurement compliance documentation.

D.7 The Escrow Engine

The escrow engine is the financial spine of the platform. Every peso that enters the platform enters the escrow engine. The engine is event-sourced: every financial event is an immutable append to the escrow_events table. The current balance is a projection of the event stream, never a mutable record. This design makes the system auditable, recoverable, and safe under Yemen network conditions.

The 2170 rule applies on the moment of funding confirmation. When a client funds a milestone, the system immediately writes three events: ESCROW_FUNDED (total amount), SPLIT_APPLIED (recording 70%, 21%, and 9% allocations), and CONTRACTOR_QUEUED (recording the contractor payment entitlement). The escrow balance displayed to the client and the talent is derived from these events in real-time.

Every write to the escrow engine carries an idempotency key — a client-generated UUID sent with the payment request. If the network drops mid-request and the client retries, the server recognises the duplicate key and returns the original result without creating a duplicate event. This single pattern is what separates a financial system from a financial system that works reliably on Yemeni mobile internet.

The 90-day reserve (21% of GPV) follows the Option C constitutional decision: at release, 50% goes back to the contractor as a quality bond release and 50% goes to the Kayan dispute fund. The contractor receives a notification: "Kayan Quality Bond — 90 Days Clean. $105 has been added to your earnings." This is a retention mechanism and a quality incentive rolled into one.

The Shariah posture: client funds sit in a segregated account at Tadamon Islamic Bank in an wadia (deposit) structure — capital-guaranteed, non-interest-bearing. Kayan is the agent, not the principal. No riba exposure. A Shariah review by an AAOIFI-certified scholar is scheduled for completion by 1 December 2026.

E · The Kayan Card System

The physical token of a digital trust architecture.

The Kayan Card is not a loyalty card and it is not a membership card. It is the physical manifestation of the platform's verification architecture — the one object a verified talent member carries that proves their standing to any client, any Hub receptionist, any café barista, and any employer they approach for a full-time role. It is a trust credential.

E.1 Four Card Types

Card	Specification
KC-T (Talent)	ISO 7810 ID-1 format (85.60mm × 53.98mm). Tier-band colour signalling: grey digital-only for T1, silver for T2, gold for T3, dark metallic for T4. Front: Kayan logo top-left, tier band as a horizontal stripe, cardholder name in Arabic (larger) and English (smaller), tier designation in Arabic, cardholder photograph (25mm × 30mm). Back: QR code (25mm × 25mm) containing the signed verification URL, issue date, expiry date, and a brief statement of the verified tier. Print: CMYK at 300 DPI with bleed, UV spot gloss on the tier band, signature strip on back.
KC-V (Volunteer)	Same physical format. Kayan sage green (569578) as the dominant colour rather than the tier band system. "كيان لليمن" embossed below the Kayan logo. Valid for 6 months from issue date. Carries the volunteer role designation and the PSEA completion date. Renewable on continued active volunteering.
KC-H (Hub Member)	Same physical format. Colour determined by membership tier (Day/Punch/Flex/Hot Desk/Dedicated/Private). Carries the membership tier, the member since date, and the access level (which Hub zones the member can access). Coterminous with the membership — expires when the membership expires.
KC-A (Kayan Forge Graduate)	Same physical format. Kayan oud-green dominant with gold elements. "Kayan Forge Graduate — خريج كيان فورج" on front. Cohort name and graduation date on back. Permanent — no expiry. The QR links to the public profile showing the capstone project summary and the cohort completion record.

E.2 The QR Verification System

Every card carries a QR code that encodes a signed URL: card.kayanwork.com/v/{card_id}?sig={signature}. The signature is an HMAC-SHA256 hash of the card ID using a secret key stored in the Kayan key management system. This prevents URL construction attacks — a forged URL will not match the signature and the verification page will show an error.

When the QR is scanned, card.kayanwork.com renders a mobile-optimised verification page in under one second. The page shows: holder name in Arabic and English, photograph, card type and tier, issue and expiry dates, and the live status indicator — green Active, red Suspended or Revoked, grey Expired. No Kayan app is required. No account is required. The page works on any smartphone with a QR scanner.

Every scan writes a card_scans event to the database: the card ID, the timestamp, the approximate scan location (from IP geolocation), and the referrer. Scan patterns are monitored: an unusual volume of scans for a single card in a short time window triggers a security alert.

The physical card is printed in batches by a DPA-signed vendor. Batches are submitted weekly from the admin console. The batch PDF is generated server-side at 300 DPI CMYK with bleed marks. The vendor ships to the Aden Hub. Every card is inspected on receipt. The cardholder signs for their card at the Hub and activates it by scanning the QR on a Hub device, confirming their identity matches the photograph.

E.3 The Venture and Startup Identification Layer

This is one of the most powerful and least obvious capabilities Kayan will build. As the platform accumulates a database of verified, track-record-rich pods and talent members, it will be able to identify pods that are exhibiting the patterns of high-potential startups: complementary micro-specializations, high team compatibility scores, exceptional delivery records, and self-directed project choices that suggest entrepreneurial orientation.

The Venture Identification module (Phase 3, from Q3 2027) will surface these pods to a dedicated section of the platform. Identified pods can opt into a "Kayan Ventures" pathway, which provides: structured mentorship from the T4 Expert Panel, introduction to Kayan's INGO and enterprise client network, and access to a Kayan-managed crowdfunding facility. The crowdfunding facility allows Kayan's community (Hub members, Kayan Forge graduates, Kayan for Yemen alumni, and verified talent) to invest directly in promising pods-turned-startups through a private crowdfunding round, followed optionally by a public round for projects with broader community benefit.

The regulatory and legal framework for the crowdfunding facility will be structured under Kayan FZ-LLC UAE, which has access to DIFC and ADGM crowdfunding frameworks. The Yemen-side projects receive the funding as a grant or soft loan through the Kayan for Yemen CSR mechanism, keeping the regulatory complexity manageable. This is not a Year 1 feature — it requires a clean database of at least 200 verified talent members with 12 months of track record. But it is designed into the data model from Day 1 so the architecture does not need to change when the feature arrives.

F · ERP, Accounting, and Automation

F.1 The Accounting Spine (Year 1)

For Year 1, the accounting system is KAYFIN_v5_Enterprise_Final.xlsx — 13 interconnected sheets, approximately 2,000 formulas, zero errors after LibreOffice recalculation. The external accountant operates from the working folder at /02_Finance_Accounting/02f_Accountant_Handover/. The FX rate is the dated Aden market rate, held in the CONFIG sheet as an editable named cell referenced by every formula in the workbook. No manual FX conversion is ever done — every transaction is posted at 1,700.

The 53-account bilingual Chart of Accounts (COA) is the foundation. Account classes: 1-1000 series for current assets including the escrow trust account, 1-2000 for fixed assets, 2-1000 for current liabilities including WHT payable and SI payable, 3-1000 for equity, 4-1000 for revenues by stream, 5-1000 for cost of revenue, 6-1000 for operating expenses including personnel, tech, and facilities, 7-1000 for finance costs, and 8-1000 for tax and zakat provisions.

F.2 The ERP Build Phases (Post-Launch)

The Excel workbook is the accounting spine through approximately Month 4. The cloud ERP build begins in Month 2 of operations to ensure the new system is live before the workbook starts to strain under growing transaction volume. Each phase is gated by the prior phase being stable for 30 days.

Phase	Module and Timeline	Key Dependency
Phase 1	Cloud GL — replace KAYFIN_v5 JOURNAL and TRIAL_BAL with a multi-user cloud GL. July 2026 to October 2026.	Clean Day-1 book with 3 months of clean journal entries. External accountant validated on COA.
Phase 2	Escrow Engine — automated 2170-rule enforcement with event sourcing on the platform ledger. September 2026 to December 2026.	Platform live with 3 months of transaction data. Shariah review complete by 1 December 2026.
Phase 3	Payroll Module — replace KAYFIN_v5 PAYROLL with an integrated payroll engine generating SI contributions, EOS accruals, and WHT calculation reports ready for YTA-Aden filing. October 2026 to January 2027.	6 months of manual payroll runs validated against the manual calculation.

Phase	Module and Timeline	Key Dependency
Phase 4	RBAC — role-based access control across all modules. Audit trail immutable log. November 2026 to March 2027.	GL and Payroll modules live and stable.
Phase 5	AI Insights Layer — variance detection, cash flow forecast, OPEX early warning, anomaly flags on escrow patterns. February 2027 to June 2027.	12 months of clean GL data as the training baseline.

F.3 Automation Layer

The automation layer runs on BullMQ. Every operational event on the platform produces one or more queue jobs. The nine queues operate continuously and independently, with their own SLAs and escalation paths. The automation layer is what allows a team of seven people to manage the operational load of what will eventually be a 100-project-per-month platform.

Examples of automation that run without human intervention: new talent registration triggers a sanctions screen job (ComplyAdvantage API call), a verification queue entry, and a welcome WhatsApp message. Milestone submission triggers a QA gate queue entry and a notification to the QA Reviewer. Escrow funding triggers the 2170 split job and a confirmation notification to all pod members. An overdue task (no update after the due date) triggers a series of escalating notifications: talent member at Day 1, Pod Lead at Day 3, Account Lead at Day 5. Review window expiry without client action triggers a reminder sequence ending in Account Lead notification.

The policy engine sits above the automation layer. It enforces rules that cannot be left to individual judgement: a project in a regulated workspace cannot have any pod member below T2; a dispute that has been open for more than 7 days must escalate to the CEO regardless of where it is in the mediation process; a talent member whose B axis drops below the T2 threshold for two consecutive 90-day cycles must have their tier reviewed by the Verification Officer. These are not manual processes — they are automated state machine transitions governed by the policy engine.

G · Infrastructure, Security, and Deployment

G.1 The Architecture in Plain Language

A user in Aden opens kayanwork.com on a phone on a 2 Mbps 3G connection. Their DNS request resolves to Cloudflare. Cloudflare's edge node in Frankfurt (closest PoP to Aden with good peering to the Middle East submarine cable) returns the cached homepage in under 500 milliseconds. If the page is not cached (a first visit after a cache clear), Cloudflare fetches from the Hetzner CX31 in Falkenstein and caches the result. The first contentful paint is under 2 seconds on this connection. That is the performance contract.

Hetzner CX31 runs Docker Compose with four containers: Next.js (three replicas behind an nginx reverse proxy for basic horizontal scaling), NestJS (three replicas), Redis (single instance with daily persistence snapshots), and MinIO (for project deliverables and card design files). PostgreSQL runs as a managed instance on Supabase (or Hetzner Managed Databases — final decision at infra setup, not a constitutional constraint) with daily WAL-based backups to AWS S3 Bahrain. Sensitive data (KYC documents, liveness videos) goes directly to AWS S3 Bahrain with KMS encryption, never touching the Hetzner disk.

Cloudflare WAF rules are tuned to the OWASP Top 10 with custom rules for the Kayan API. Rate limiting: 60 requests per minute per IP on public pages, 120 per minute per authenticated user on the platform, 300 per minute for admin users. The escrow API endpoints have the most aggressive rate limiting: 10 requests per minute per authenticated user. Every rate limit hit generates a Sentry event.

G.2 Security Model

Control	Implementation
Authentication	Auth.js v5 magic-link primary. SMS OTP via Twilio for users without reliable email. TOTP MFA (Google Authenticator-compatible) required for admin and finance roles. Session tokens are JWTs signed with RS256, rotated every 14 days or on privilege change.
Authorisation	PostgreSQL row-level security (RLS) enforces that every query returns only rows the authenticated user is permitted to see. NestJS guards enforce route-level RBAC. The RBAC matrix has seven roles: Client, Talent, Pod Lead, QA Reviewer, Account Lead, Finance, and Admin. Each role is least-privilege by default — any permission grant requires a documented justification in the RBAC policy.
Data Encryption	In transit: TLS 1.3 everywhere, enforced by Cloudflare. At rest: AES-256 on all database volumes. KYC data: AES-256 via AWS KMS with the key managed in a separate AWS region from the

Control	Implementation
	data.
Audit Trail	Every privileged action (payment approval, tier change, admin override, dispute resolution) writes an immutable audit event. The audit log table has no UPDATE or DELETE permissions at the database level — only INSERT is allowed from the application service account.
Code Security	GitHub Dependabot for dependency updates. npm audit on every CI run. No secrets in the codebase — all secrets in environment variables pulled from a Hetzner-hosted Vault instance at runtime. Regular (quarterly) penetration test by an external security consultant.
Data Residency	User PII: PostgreSQL on Hetzner Germany. Biometric and KYC data: AWS S3 Bahrain. No user data is stored in the United States or processed by US-jurisdiction services unless the user is a US-based client (handled case by case via the legal layer).
Incident Response	Sentry for error tracking. PagerDuty-equivalent on-call rotation (Phase 2 — Phase 1 is WhatsApp alerts to the dev lead). Data breach: the 72-hour GDPR-shaped breach response runbook (KAY-RLC-011) applies regardless of Yemen's absence of a data protection law. Incident post-mortems are published on status.kayanwork.com for all P1 incidents.

G.3 Deployment and CI/CD

The deployment pipeline runs on GitHub Actions. On every push to the main branch: Prisma migrations run against a test database, the NestJS test suite runs, the Next.js build runs. If all three pass, the deployment job SSHs into Hetzner and runs docker compose pull && docker compose up -d --no-deps app. Zero manual steps. The entire pipeline completes in under 8 minutes. Failed deployments automatically trigger a rollback to the previous image.

Environment management: three environments. Development runs on a developer's local machine with Docker Compose. Staging runs on a dedicated Hetzner CX21 instance and receives every push to the develop branch. Production runs on Hetzner CX31 and receives only tagged releases that have been in staging for at least 24 hours without a P1 error.

Database migrations follow a zero-downtime migration strategy: every migration is backwards-compatible with the previous application version. This means that during a deployment, the old version of the application and the new database schema can coexist without errors. Irreversible schema changes (column drops, type changes) require a three-phase migration: add the new column, migrate the data, drop the old column — each phase in a separate deployment.

H · Build Plan, Accountability, and What Gets Built When

H.1 Phase 0 (Before 25 April 2026)

volunteers.kayanwork.com is the first production surface. It goes live on 25 April 2026 and is the simplest system in the Kayan ecosystem: a static Next.js page with a server-side form submission to PostgreSQL. No authentication required. No payment. No matching. Just: a volunteer application form that saves to the database, sends a WhatsApp notification to the Kayan for Yemen Coordinator, and displays a confirmation to the applicant. Plus the static kayanwork.com homepage — the brand surface. Total build time for Phase 0: 3-4 days for an experienced Next.js developer.

H.2 Phase 1 (1 May to 30 June 2026 — the launch sprint)

Phase 1 builds everything required for the 1 June commercial launch. The build is sequenced so the most critical paths are completed first: the verification queue and talent registration before the matching engine, the matching engine before the client brief flow, the client brief flow before the escrow, the escrow before the workspace.

Week	Build Target
Week 1 (1-7 May)	Hetzner provisioned. DNS configured. Cloudflare WAF deployed. Auth.js magic-link working. PostgreSQL + Prisma schema v1 deployed. NestJS API skeleton: /users, /auth routes. kayanwork.com static site live (all public pages including Manifesto, About, Team, Verified Standard, Press, Careers, Contact, and six legal pages).
Week 2 (8-14 May)	Talent registration flow: /talent/register — profile creation, specialization selection, portfolio upload, consent capture. Verification queue in admin console. Identity verification appointment booking (Calendly integration). Sanctions screening job (ComplyAdvantage API). Card issuance workflow backend.
Week 3 (15-21 May)	Client registration and brief intake wizard. Claude SOW generation (API integration + prompt). Matching engine: Cohere embedding, pgvector retrieval, Claude Haiku reranking. Pod assembly flow and client approval. Escrow funding via Stripe. 2170 split job in BullMQ. space.kayanwork.com Hub booking via Spacebring integration.
Week 4 (22-31 May)	E-working space: all eight modules. G5 to G9 gate transitions. QA gate queue. Payout release jobs. Admin console: all nine queues operational. forge.kayanwork.com cohort registration. impact.kayanwork.com initial dashboard. status.kayanwork.com.

Week	Build Target
	card.kayanwork.com QR verification. Full end-to-end test: client posts brief → pod assembled → escrow funded → milestone delivered → payout released. Zero errors.

H.3 Phase 2 (June to September 2026)

Phase 2 adds the marketplace mode (Mode B), the yeto.kayanwork.com Ready Products catalogue, the Kayan Card print management system (batch generation, vendor transmission, QR activation), the Kayan Forge Moodle integration with the platform (completion webhook → T axis update → KC-A issuance), the bi-directional API for enterprise clients, and the impact dashboard expanded with SDG-aligned metrics pulled from the talent and escrow databases.

H.4 Phase 3 (October to December 2026)

Phase 3 adds: the ERP cloud GL (replacing KAYFIN_v5 JOURNAL and TRIAL_BAL), the automated payroll module, the enterprise API at api.kayanwork.com/v1, the developer documentation at docs.kayanwork.com, the subscription/retainer commercial mode (Mode C), the Kayan Card partner licensing endpoint, and the first version of the AI Insights early-warning dashboard.

H.5 Who Owns What

Responsibility	Owner	Escalation Path
All platform architecture decisions	Dev Lead (Mustafa Haney, Cairo)	CEO on any architectural change
All security decisions	Dev Lead with CEO sign-off	External security consultant quarterly
Database schema changes	Dev Lead	CEO reviews any migration touching financial tables
API endpoint design	Dev Lead	CEO reviews any endpoint touching payment or PII
Third-party service selection	Dev Lead proposes, CEO approves	Board for any contract above $500/month
Deployment to production	Dev Lead (automated pipeline)	CEO approves any manual deployment override

Responsibility	Owner	Escalation Path
Egypt contractor work	Dev Lead manages and reviews	NDA signed before any work begins
All source code	GitHub org owned by Kayan Innovation LLC	CEO has org owner access at all times
Tech OS updates	Dev Lead proposes, CEO approves and signs	Archived in KAY-ARC when superseded

H.6 The 14 Things That Must Be True on 30 June 2026

This is the go/no-go checklist. If any of these 14 conditions is false at the start of 30 June 2026, the launch does not proceed.

- All 14 subdomains return a 200 response in Arabic and English.
- The full client journey (brief → SOW → pod → escrow → workspace → payout) completes end-to-end in a test transaction with zero errors.
- All conversion events (brief_submitted, verification_approved, escrow_funded, milestone_approved) write correctly to the audit log.
- ComplyAdvantage sanctions screening fires on every new user registration.
- The Stripe payment flow completes end-to-end in a test transaction of $10.
- The 2170 split fires correctly: $7.00 to contractor queue, $2.10 to reserve, $0.90 to fee on a $10 test transaction.
- The QR verification endpoint on card.kayanwork.com renders in under 1 second on a simulated 2 Mbps connection.
- WhatsApp Business notifications fire for: new talent registration, verification appointment confirmation, SOW ready for review, escrow funded, milestone approved.
- status.kayanwork.com shows all subdomains green.
- Cloudflare WAF rules deployed and tested against common injection payloads.
- PostgreSQL daily backup verified by restoring to a clean database and confirming data integrity.
- KAYFIN_v5 Excel workbook has zero formula errors after LibreOffice recalculation and is in the accountant handover folder.
- At least 20 talent members are verified at T2 Silver with KC-T cards ready for collection on Day 1.
- At least one client brief has been posted, matched, and scoped during the pre-launch period so the first paying project can begin on Day 1.

KAY-TECH-OS-008 v8.0 — Final. Ratified: 20 April 2026. Owner: CEO + Dev Lead Mustafa Haney.

Stack locked under ADR-001. Next scheduled review: October 2026.